

INTELLIGENT SDN-BASED TRAFFIC CLASSIFICATION AND QOS CONTROL FOR SMART CLASSROOM NETWORKS

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ABSTRACT

The integration of digital technologies into modern education has transformed traditional classrooms into smart learning environments where students rely heavily on internet-connected devices for accessing academic resources. While this transformation enhances learning experiences, it also introduces challenges such as the misuse of network resources for non-academic activities including social media browsing, video streaming, and online entertainment. These activities not only reduce student engagement but also lead to inefficient utilization of network bandwidth, affecting the performance of critical educational services.

This research proposes an intelligent Software-Defined Networking (SDN)-based system for real-time traffic classification and Quality of Service (QoS) enforcement in smart classroom environments. The system is implemented using the Ryu SDN controller and OpenFlow protocol. Traffic classification is performed using a supervised machine learning model based on flow-level statistical features, Domain Name System (DNS) information, and Transport Layer Security (TLS) Server Name Indication (SNI).

The system classifies traffic as academic or non-academic and dynamically applies QoS policies. Academic traffic is prioritized using Differentiated Services Code Point (DSCP) marking, while non-academic traffic is rate-limited using OpenFlow meter tables. The design avoids deep packet inspection, ensuring compatibility with encrypted traffic and preserving user privacy.

The proposed system provides a scalable, efficient, and privacy-aware solution for improving network performance and enhancing learning environments in smart classrooms.

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CHAPTER 1 — INTRODUCTION

1.1 Background Literature

The rapid advancement of information and communication technologies (ICT) has significantly transformed the education sector, leading to the emergence of smart classroom environments where digital tools and internet connectivity play a central role in teaching and learning processes. In these environments, students rely heavily on internet-enabled devices such as laptops, tablets, and smartphones to access academic resources, participate in online lectures, and collaborate through cloud-based platforms. This transformation has improved accessibility to knowledge and enabled flexible and interactive learning experiences [1].

However, the widespread availability of internet access in classrooms has also introduced several challenges. One of the most critical issues is the misuse of network resources by students for non-academic purposes. Studies indicate that students frequently engage in activities such as social media browsing, video streaming, online gaming, and entertainment during class hours, which negatively affects their concentration and academic performance [2]. In addition to behavioral impacts, these activities consume a significant portion of available network bandwidth, leading to congestion and reduced performance of essential educational applications such as video conferencing systems and learning management platforms [3].

Traditional network management approaches, including access control lists (ACLs), firewall filtering, and port-based traffic identification, are not effective in modern network environments. These methods rely on static configurations and cannot adapt to dynamic traffic patterns. Furthermore, the increasing use of encryption protocols such as HTTPS and DNS-over-HTTPS limits visibility into packet contents, making it difficult to classify traffic using conventional techniques. Deep Packet Inspection (DPI), although effective in some cases, is computationally expensive and raises privacy concerns [4].

Software-Defined Networking (SDN) has emerged as a promising solution for addressing these challenges. SDN introduces a paradigm shift by separating the control plane from the data plane, enabling centralized and programmable network management. This allows network administrators to monitor traffic in real time and enforce policies dynamically based on network conditions [5]. By integrating machine learning techniques with SDN, it becomes possible to classify network traffic based on behavioral characteristics rather than static rules. This approach enables accurate identification of academic and non-academic traffic even in encrypted environments, providing a scalable and efficient solution for smart classroom network management.

1.2 Research Gap

Despite the extensive research conducted in the fields of network traffic classification and Software-Defined Networking, several limitations remain in existing approaches. Traditional traffic classification methods such as port-based identification and deep packet inspection are no longer effective in modern networks. Port-based methods fail due to the use of dynamic ports by applications, while deep packet inspection is limited by encryption and raises privacy concerns [6].

Recent studies have explored machine learning-based approaches for traffic classification. These methods use statistical features such as packet size, flow duration, and inter-arrival time to identify patterns in network traffic. While these approaches have shown promising results, many of them are evaluated using offline datasets or simulated environments. As a result, their performance often does not accurately reflect real-world conditions, where traffic is dynamic, encrypted, and influenced by factors such as Content Delivery Networks (CDNs).

Another significant limitation in existing research is the lack of integration between traffic classification and Quality of Service (QoS) enforcement. Many studies focus solely on classification without implementing mechanisms to control traffic based on classification results. In cases where QoS is implemented, it is often static and does not adapt to real-time network conditions.

Furthermore, there is limited research specifically focused on smart classroom environments. Most studies address general network scenarios without considering the unique requirements of educational settings, such as prioritizing academic traffic while restricting non-academic usage. Additionally, many existing solutions do not fully utilize the capabilities of SDN controllers for real-time decision-making and dynamic policy enforcement.

These limitations highlight a clear research gap for an integrated system that combines SDN and machine learning to perform real-time traffic classification and dynamic QoS enforcement in realistic and encrypted network environments.

1.3 Research Problem

The primary problem addressed in this research is the inability of existing classroom network infrastructures to effectively manage and control network traffic based on its purpose. In most educational environments, all network traffic is treated equally, regardless of whether it is related to academic activities or not. This results in inefficient bandwidth utilization and negatively impacts the performance of critical educational applications.

One of the key challenges in solving this problem is the accurate classification of network traffic. Modern network traffic is highly dynamic and often encrypted, making it difficult to analyze using traditional methods. Techniques such as deep packet inspection are not feasible due to privacy concerns and high computational requirements, while port-based classification is unreliable due to dynamic port usage.

Additionally, the use of Content Delivery Networks (CDNs) complicates traffic identification, as multiple services may share the same IP addresses. This makes it difficult to determine the nature of traffic based solely on IP-level analysis. DNS-over-HTTPS further reduces visibility into domain-level information, making classification even more challenging.

As a result, there is a need for an intelligent system that can classify network traffic accurately without relying on payload inspection. The system must operate in real time, adapt to changing traffic patterns, and be capable of handling encrypted traffic. Furthermore, it should enforce appropriate QoS policies to prioritize academic traffic and limit non-academic usage, ensuring efficient network performance in smart classroom environments.

1.4 Research Objectives

The main objective of this research is to design and implement an intelligent network management system that integrates Software-Defined Networking (SDN) with machine learning techniques to classify network traffic and enforce Quality of Service (QoS) policies in real time.

The specific objectives of this research are as follows:

To analyze the limitations of existing traffic classification and network management techniques in modern educational environments

To design an SDN-based architecture that enables centralized monitoring and control of network traffic

To develop a machine learning model capable of classifying network traffic into academic and non-academic categories using flow-level features

To integrate the machine learning model with an SDN controller for real-time traffic classification and decision-making

To implement dynamic QoS policies that prioritize academic traffic and limit non-academic usage using mechanisms such as DSCP marking and rate limiting

To evaluate the performance of the proposed system under realistic network conditions, including encrypted traffic and CDN-based routing

To identify the limitations of the system and propose potential improvements for future research

1.5 Scope of the Study

This research focuses on the development of a flow-based traffic classification and QoS enforcement system specifically designed for smart classroom environments. The system operates in real time and analyzes network traffic using statistical and domain-level features rather than packet payloads. This approach ensures compatibility with encrypted traffic and preserves user privacy.

The study considers a network setup where multiple student devices are connected through an SDN-enabled switch, and all traffic is routed through a centralized SDN controller. The classification process is based on features such as packet size, inter-arrival time, flow duration, packet count, and byte count. Domain-level information is obtained using DNS queries and TLS Server Name Indication (SNI) to enhance classification accuracy.

The research is limited to binary classification, where traffic is categorized as either academic or non-academic. It does not include multi-class classification or detailed application-level identification. Additionally, the system does not perform deep packet inspection or decrypt encrypted traffic, as these approaches are beyond the scope of this study.

The QoS implementation focuses on prioritization and rate limiting using OpenFlow mechanisms such as DSCP marking and meter tables. Advanced network security features such as intrusion detection systems and firewall integration are not considered within the scope of this research.

1.6 Significance of the Study

The significance of this research lies in its ability to address a critical challenge in modern educational environments: efficient management of network resources in the presence of diverse and dynamic traffic. By providing an intelligent system for traffic classification and QoS enforcement, this research contributes to improving the overall performance and reliability of smart classroom networks.

The proposed system ensures that academic traffic receives priority, thereby enhancing the performance of essential educational applications such as video lectures, online assessments, and collaborative tools. This leads to improved learning experiences for students and better teaching outcomes for instructors.

Another important contribution of this research is its privacy-preserving approach. By avoiding deep packet inspection and relying on flow-level features, the system respects user privacy while still achieving effective classification. This makes it suitable for deployment in real-world environments where privacy concerns are a major consideration.

Furthermore, the integration of SDN and machine learning demonstrates the potential of combining emerging technologies to solve complex networking problems. The centralized control provided by SDN enables dynamic policy enforcement, while machine learning enables intelligent decision-making based on traffic patterns. This approach provides a scalable and adaptable solution that can be extended to other domains such as enterprise networks and public internet infrastructure.

1.7 Chapter Summary

This chapter presented a comprehensive introduction to the research by discussing the background, research gap, problem statement, objectives, scope, and significance of the study. It highlighted the transformation of traditional classrooms into smart learning environments and identified the challenges associated with unrestricted internet usage and inefficient network management.

The chapter also examined the limitations of existing traffic classification techniques and emphasized the need for an integrated solution that combines SDN and machine learning. The research problem was clearly defined, focusing on the inability of current systems to differentiate between academic and non-academic traffic and enforce QoS policies effectively.

The objectives of the study were outlined to provide a clear direction for the research, while the scope and significance were defined to establish the relevance and impact of the proposed system. Overall, this chapter provides a strong foundation for the subsequent chapters, which will explore existing literature, describe the proposed methodology, and present the implementation and evaluation of the system.

CHAPTER 2 - LITERATURE REVIEW

2.1 Introduction

The rapid proliferation of internet-based applications and the increasing reliance on digital platforms in educational environments have significantly complicated modern network management. Smart classrooms, which integrate online learning systems, multimedia content delivery, and collaborative tools, depend heavily on stable and efficient network performance. However, these environments also experience a mixture of academic and non-academic traffic generated by students and staff, resulting in competition for limited bandwidth and potential degradation of service quality for critical educational applications [3].

The coexistence of diverse traffic types introduces challenges such as congestion, latency, and unfair bandwidth allocation. Non-academic activities—including social media usage, video streaming, and online gaming—often consume disproportionate bandwidth, adversely affecting the performance of essential services like virtual classrooms and cloud-based learning platforms. Consequently, accurate traffic classification and effective Quality of Service (QoS) enforcement are necessary to ensure that academic activities receive priority and consistent performance [8].

Traditional network management techniques struggle to cope with modern traffic characteristics. The widespread adoption of encryption protocols such as HTTPS and DNS-over-HTTPS has reduced visibility into packet contents, making conventional approaches like port-based classification and deep packet inspection less effective [4]. These limitations highlight the need for advanced methods capable of operating in encrypted and dynamic environments.

Software-Defined Networking (SDN) provides a promising solution by enabling centralized control, programmability, and real-time network management [5]. When combined with machine learning techniques, SDN can support intelligent traffic classification and adaptive QoS enforcement. This chapter reviews the key concepts and existing approaches related to SDN, OpenFlow, traffic classification, machine learning, and QoS, and identifies the limitations that motivate the proposed system.

2.2 Software-Defined Networking (SDN)

Software-Defined Networking (SDN) represents a paradigm shift in network architecture by decoupling the control plane from the data plane, thereby enabling centralized and programmable network management [5]. In conventional networks, each device independently determines how to forward packets based on locally stored rules, which complicates configuration and limits adaptability. SDN addresses this by introducing a logically centralized controller that governs network behavior through software-defined policies.

The SDN architecture is typically organized into three layers: the application layer, control layer, and data plane. The application layer hosts network applications that define high-level policies, such as routing, security, and QoS requirements. The control layer consists of the SDN controller, which translates these policies into specific instructions. The data plane comprises forwarding devices, such as switches, that execute these instructions to handle packet forwarding [1], [2].

A key advantage of SDN is its global network visibility. Because the controller maintains a comprehensive view of network state, it can make informed decisions regarding routing, congestion management, and resource allocation. This is particularly beneficial in smart classroom environments, where traffic patterns are dynamic and user behavior varies throughout the day.

SDN also enables rapid reconfiguration. Network policies can be updated centrally and propagated instantly across all devices, reducing operational overhead and minimizing downtime. Additionally, SDN supports automation, allowing administrators to implement complex policies using software logic rather than manual configuration.

Despite these advantages, SDN introduces challenges related to scalability and reliability. A centralized controller can become a bottleneck under heavy traffic loads or a single point of failure if redundancy is not implemented. Techniques such as distributed control planes and controller replication are commonly used to mitigate these issues [5]. Overall, SDN provides the flexibility and control required for implementing intelligent traffic management systems in modern networks.

2.3 OpenFlow Protocol

OpenFlow is a foundational protocol in SDN that facilitates communication between the controller and network switches, enabling centralized control over packet forwarding behavior [6]. Through OpenFlow, the controller installs flow rules in switches, dictating how packets matching specific criteria should be processed.

Each OpenFlow-enabled switch maintains one or more flow tables. A flow table entry typically consists of match fields, actions, counters, and priority values. Match fields specify packet attributes such as source/destination IP addresses, transport ports, and protocol types. Actions determine how packets should be handled—such as forwarding to a port, dropping, or modifying header fields—while priorities resolve conflicts when multiple rules match a packet.

When a packet arrives at a switch, it is matched against existing flow entries. If a matching rule is found, the switch executes the associated action locally. If no match exists, the switch generates a Packet-In message and forwards the packet (or its header) to the controller for decision-making. The controller then analyzes the packet and installs a corresponding flow rule, allowing subsequent packets of the same flow to be processed without controller intervention. This hybrid reactive–proactive behavior reduces latency over time and improves scalability [5], [6].

OpenFlow also includes advanced constructs such as meter tables and group tables. Meter tables enable rate control (e.g., policing and shaping), which is essential for implementing QoS policies like bandwidth limiting. Group tables support more complex forwarding behaviors such as load balancing and fast failover. These capabilities make OpenFlow particularly suitable for dynamic traffic control in SDN-based systems.

In the context of this research, OpenFlow provides the mechanism to enforce classification outcomes—e.g., prioritizing academic flows or rate-limiting non-academic flows—by installing appropriate rules and meters at the switch level.

2.4 Traffic Classification Techniques

Traffic classification is a core function in network management, enabling the identification of application types and supporting policy enforcement such as QoS and access control [4]. As network traffic has evolved—becoming more encrypted, dynamic, and application-agnostic—classification techniques have also progressed from simple heuristics to data-driven methods.

2.4.1 Port-Based Classification

Port-based classification is one of the earliest and simplest approaches, relying on well-known transport-layer port numbers to infer application types. For example, HTTP typically uses port 80 and HTTPS uses port 443. Because it only inspects header fields, this method is computationally efficient and easy to deploy.

However, its effectiveness has diminished significantly in modern networks. Many applications use dynamic or ephemeral ports, and some deliberately tunnel their traffic over common ports to evade filtering. As a result, the same port can carry multiple application types, and a single application can use multiple ports. This ambiguity leads to poor classification accuracy and makes port-based methods unreliable for contemporary traffic analysis [4].

Furthermore, port-based classification provides no insight into traffic behavior. It cannot distinguish between different uses of the same application (e.g., educational vs. entertainment content over HTTPS), which is essential in smart classroom contexts. Consequently, while still used for coarse filtering, port-based classification is insufficient as a standalone solution.

2.4.2 Deep Packet Inspection (DPI)

Deep Packet Inspection (DPI) analyzes packet payloads to identify application signatures, offering fine-grained visibility into network traffic. By matching payload content against known patterns, DPI can accurately classify applications and even detect specific services or protocols encapsulated within others.

Despite its accuracy, DPI faces several practical limitations. First, it is computationally intensive, requiring significant processing resources to inspect payloads at line rate particularly in high-throughput networks. This makes real-time deployment challenging without specialized hardware.

Second, DPI raises privacy concerns because it involves inspecting user data. In environments such as educational institutions, this can conflict with privacy policies and regulations. Third, and most critically, the widespread adoption of encryption (e.g., HTTPS, TLS 1.3) renders payloads inaccessible, severely limiting DPI's effectiveness. Encrypted traffic prevents signature matching, reducing DPI to metadata analysis, which undermines its primary advantage [4].

Given these constraints, DPI is less suitable for modern, privacy-sensitive, and encrypted network environments, motivating the shift toward alternative techniques.

2.4.3 Statistical and Flow-Based Classification

Statistical and flow-based classification techniques focus on traffic behavior rather than content. They extract features such as packet size distributions, inter-arrival times, flow duration, packet counts, and byte volumes to characterize traffic patterns. Packets are aggregated into flows, typically defined by a 5-tuple (source IP, destination IP, source port, destination port, protocol) and analyzed over time.

This approach offers several advantages. First, it is payload-agnostic, making it effective for encrypted traffic. Second, it is scalable, as it processes flows rather than individual packets, reducing computational overhead. Third, it captures behavioral signatures of applications; for example, streaming traffic often shows steady bitrates and regular packet intervals, while web browsing exhibits bursty, short-lived flows [3].

However, statistical methods can face ambiguity when different applications exhibit similar patterns (e.g., certain streaming and conferencing behaviors). They may also require an observation window to collect sufficient data, introducing slight delays before confident classification can be made. Despite these limitations, flow-based methods form a strong foundation for modern classification and are often combined with machine learning to improve accuracy.

2.4.4 Machine Learning-Based Classification

Machine learning (ML) has emerged as a powerful approach for traffic classification, leveraging data-driven models to learn complex patterns from flow-level features [4]. In supervised learning, models are trained on labeled datasets where each flow is tagged (e.g., academic vs. non-academic). The trained model then predicts labels for new, unseen flows.

Common algorithms include Logistic Regression, Decision Trees, Support Vector Machines, and ensemble methods. Among these, Logistic Regression is particularly attractive for real-time systems due to its simplicity, interpretability, and low computational cost. It models the probability of a class using a linear combination of features and a sigmoid function, enabling fast inference suitable for controller integration [7].

ML-based methods are adaptive and can generalize to unseen traffic patterns better than rule-based approaches. They also operate on statistical features, making them compatible with encrypted traffic. However, their performance depends on data quality, feature engineering, and model generalization. Poorly labeled datasets or biased samples can degrade accuracy. Additionally, concept drift (changes in traffic patterns over time) may require periodic retraining.

In practice, combining flow-based features with ML models provides a robust and scalable solution for traffic classification in modern networks.

2.5 Machine Learning in Networking

Machine learning is increasingly applied across networking tasks, including traffic classification, anomaly detection, congestion prediction, and resource optimization. Its strength lies in identifying non-linear relationships and subtle patterns that are difficult to capture with static rules.

In traffic classification, supervised learning dominates due to the availability of labeled datasets. Models learn mappings from feature vectors (e.g., flow statistics) to class labels. Logistic Regression is widely used for binary classification tasks because it is efficient, stable, and easy to deploy within SDN controllers for near real-time inference [7].

While deep learning models (e.g., CNNs, RNNs) can achieve higher accuracy by learning hierarchical representations, they often require significant computational resources and large datasets. This can limit their practicality in controller-based deployments where latency and resource usage are critical.

An important consideration is feature selection. Features must be informative, robust to encryption, and inexpensive to compute. Combining statistical features with contextual signals (e.g., DNS or TLS SNI when available) can improve performance. Additionally, model evaluation should consider not only accuracy but also latency, resource overhead, and robustness to unseen traffic.

Integrating ML with SDN enables closed-loop control: the controller classifies traffic and immediately enforces policies, creating an adaptive and intelligent network management system.

2.6 Quality of Service (QoS)

Quality of Service (QoS) mechanisms ensure that different traffic classes receive appropriate treatment based on their importance, thereby maintaining network performance under contention [8]. In smart classroom environments, QoS is essential to prioritize academic applications over non-academic ones.

Key QoS techniques include:

- Traffic shaping and policing: controlling the rate at which traffic is sent or allowed.
- Priority queuing: serving higher-priority traffic first to reduce latency and jitter.
- Differentiated Services (DiffServ) and DSCP marking: labeling packets with priority levels so downstream devices can handle them accordingly.
- Rate limiting via meters: enforcing bandwidth caps on selected flows.

In SDN, QoS policies can be applied dynamically through the controller. For instance, the controller can mark academic traffic with higher DSCP values and install meter rules to limit non-academic flows. This dynamic enforcement allows the network to adapt to changing conditions and maintain service quality for critical applications [8].

A key design consideration is balancing fairness and performance: while non-academic traffic should be controlled, it should not be completely blocked unless policy requires it. Proper QoS configuration ensures equitable resource allocation while safeguarding the performance of essential services.

2.7 Challenges in Modern Network Traffic Classification

Modern networks present several challenges that complicate traffic classification. Encryption is the most significant, as protocols like HTTPS and TLS 1.3 conceal payload data, limiting visibility for content-based methods [4]. Even metadata can be reduced with techniques like encrypted SNI (ESNI), further restricting domain-level insights.

DNS-over-HTTPS (DoH) encrypts DNS queries, preventing straightforward domain resolution mapping. Content Delivery Networks (CDNs) distribute services across shared IP infrastructures, causing multiple applications to resolve to the same IP addresses, which undermines IP-based identification.

Additionally, application multiplexing and protocol obfuscation blur traffic signatures, while concept drift—changes in traffic behavior over time—can degrade model performance if not addressed through retraining or adaptive methods.

These challenges necessitate hybrid approaches that combine statistical features, available contextual signals (e.g., DNS/SNI when present), and machine learning models capable of generalizing beyond fixed signatures.

2.8 Chapter Summary

This chapter reviewed the state of the art in Software-Defined Networking, OpenFlow, traffic classification techniques, machine learning applications, and Quality of Service mechanisms. It highlighted the limitations of traditional methods—particularly in the face of encryption and dynamic application behavior—and emphasized the strengths of flow-based and machine learning approaches.

The analysis shows that integrating SDN with machine learning enables real-time, adaptive traffic classification and dynamic QoS enforcement, making it well-suited for smart classroom environments. These insights inform the design choices in the next chapter, which presents the methodology of the proposed system.

CHAPTER 3 — METHODOLOGY

3.1 Introduction

This chapter presents the methodology used to design and implement the proposed intelligent traffic classification and Quality of Service (QoS) enforcement system for smart classroom environments. The methodology integrates Software-Defined Networking (SDN) with machine learning techniques to enable real-time monitoring, classification, and control of network traffic. The design is motivated by the limitations of traditional traffic classification approaches, particularly their inability to handle encrypted traffic and dynamic network behavior effectively [4].

The proposed system adopts a flow-based traffic analysis approach combined with supervised machine learning. Unlike Deep Packet Inspection (DPI), which relies on payload inspection, this system uses statistical and contextual features, making it suitable for modern encrypted environments [9]. The SDN controller plays a central role in collecting network data, performing classification, and enforcing QoS policies dynamically [5].

This chapter provides a detailed description of the system architecture, data collection process, feature extraction techniques, machine learning model design, QoS policy implementation, and system workflow. The methodology is designed to be scalable, privacy-preserving, and suitable for real-world deployment.

3.2 System Architecture

The proposed system is based on the SDN architecture, which separates the control plane from the data plane to enable centralized network management [5]. The architecture consists of three main components: end-user devices, an SDN-enabled switch, and an SDN controller.



End-user devices include laptops, smartphones, and tablets used by students in a classroom environment. These devices generate both academic and non-academic traffic, which is transmitted through a wireless access point and forwarded to an SDN-enabled switch such as Open vSwitch.

The SDN switch acts as the data plane, forwarding packets based on flow rules installed by the controller. The controller, implemented using the Ryu framework, represents the control plane and is responsible for traffic monitoring, feature extraction, classification, and QoS enforcement.

Communication between the controller and the switch is established using the OpenFlow protocol, which allows dynamic installation of flow rules [6]. The centralized nature of SDN enables the controller to maintain a global view of the network, facilitating intelligent decision-making and efficient resource allocation [10].

3.3 System Workflow

The system operates through a structured workflow that enables real-time traffic classification and control. When a packet arrives at the SDN switch, the switch checks its flow table for a matching rule. If no rule is found, the packet is forwarded to the controller as a Packet-In message.

The controller extracts header information such as IP addresses, port numbers, and protocol type. Packets are grouped into flows based on these attributes. Flow statistics are continuously updated as new packets arrive.

Once sufficient data is collected for a flow, feature extraction is performed. The extracted features are passed to the trained machine learning model, which predicts whether the traffic is academic or non-academic.

Based on the prediction result, the controller installs flow rules in the switch. These rules define how subsequent packets of the flow should be handled, including prioritization or rate limiting. This reduces controller overhead and improves system efficiency.

3.4 Flow-Based Traffic Modeling

The system uses flow-based traffic modeling, where a flow is defined as a sequence of packets sharing common attributes such as source IP, destination IP, source port, destination port, and protocol [3].

Flow-based analysis provides several advantages over packet-level analysis. It reduces computational complexity by processing groups of packets instead of individual packets. It also captures temporal behavior, which is essential for distinguishing between different types of applications.

Each flow is tracked using a unique identifier, and statistics such as packet count, byte count, and timestamps are recorded. These statistics are used to extract features that represent the behavior of the flow.

Flow-based modeling is particularly effective in encrypted environments, as it does not rely on payload inspection. Instead, it uses observable traffic characteristics, making it suitable for modern network environments [9].

3.5 Feature Extraction

Feature extraction is a critical component of the classification process. The system extracts both statistical and contextual features from each flow to capture its behavior. Statistical features include packet size, flow duration, inter-arrival time, packet count, and byte count. These features provide insights into traffic patterns and help differentiate between application types [3].

Flow duration is calculated as the time difference between the first and last packets in a flow. Inter-arrival time measures the time between consecutive packets, which helps identify traffic patterns such as streaming or browsing behavior.

Contextual features are extracted using DNS queries and TLS Server Name Indication (SNI). DNS analysis provides domain names associated with IP addresses, while SNI reveals the intended server name during encrypted connections. These features enhance classification accuracy without violating privacy. Feature normalization is performed to ensure consistency and improve model performance. The final feature vector is used as input to the machine learning model.

3.6 Machine Learning Model

The proposed system uses a supervised machine learning approach for traffic classification. Logistic Regression is selected as the primary model due to its efficiency, interpretability, and suitability for real-time applications [7].

The model is trained using a labeled dataset containing flow features and corresponding labels (academic or non-academic). During training, the model learns the relationship between input features and output classes.

Logistic Regression estimates the probability of a flow belonging to a particular class using the sigmoid function. The model is defined as:

$$y = \frac{1}{1+e^{-(w_0 + w_1 x)}}$$

where x represents the feature vector, w_1 represents model weights, and w_0 is the bias term.

The trained model is serialized and integrated into the SDN controller. During runtime, the controller uses the model to classify incoming flows in real time.

3.7 QoS Policy Design

Quality of Service (QoS) policies are implemented to prioritize academic traffic and limit non-academic usage. The system uses DSCP marking and OpenFlow meter tables to enforce QoS policies [8].

Academic traffic is assigned a higher priority by modifying the DSCP field in the IP header. This ensures that academic packets are processed with higher priority in the network.

Non-academic traffic is rate-limited using meter tables. A predefined bandwidth limit is applied to such traffic, preventing it from consuming excessive resources. This ensures fair bandwidth allocation and improves network performance.

QoS policies are dynamically applied based on classification results, allowing the system to adapt to changing traffic conditions.

3.8 Decision Logic and Classification Timing

To improve classification accuracy, the system uses an observation window before making a classification decision. Instead of classifying traffic based on a single packet, the system collects multiple packets belonging to the same flow.

The observation window is defined by a minimum number of packets or a time threshold. Once this threshold is reached, feature extraction and classification are performed.

This approach ensures that sufficient data is available for accurate classification and reduces the likelihood of incorrect predictions.

3.9 Integration with SDN Controller

The integration of the machine learning model with the SDN controller enables real-time traffic classification and control. The controller handles all operations, including packet processing, feature extraction, classification, and policy enforcement.

When a Packet-In message is received, the controller updates flow statistics and checks whether the observation window is complete. If so, it performs classification and installs flow rules accordingly.

This integration allows seamless operation and ensures that classification decisions are enforced immediately in the network.

3.10 Testing Methodology

The system is evaluated using a structured testing methodology. Test cases are designed to simulate real-world traffic scenarios, including academic and non-academic applications.

Performance metrics such as classification accuracy, latency, and bandwidth utilization are measured. The results are used to evaluate system performance and identify limitations.

3.11 Chapter Summary

This chapter presented the methodology used to design and implement the proposed system. It described the system architecture, workflow, feature extraction process, machine learning model, QoS policy design, and integration with the SDN controller.

The methodology demonstrates how SDN and machine learning can be combined to achieve intelligent traffic classification and dynamic QoS enforcement. The next chapter presents the implementation and evaluation of the system.

CHAPTER 4 — IMPLEMENTATION AND SYSTEM EVALUATION

4.1 Introduction

This chapter presents the detailed implementation and evaluation of the proposed intelligent traffic classification and Quality of Service (QoS) enforcement system designed for smart classroom environments. The system integrates Software-Defined Networking (SDN) with machine learning techniques to enable real-time traffic monitoring, classification, and dynamic bandwidth control.

Unlike traditional network management approaches, the proposed system is implemented in a real-time SDN environment where all network traffic is centrally controlled and analyzed. The implementation focuses on handling modern challenges such as encrypted traffic, dynamic application behavior, and bandwidth contention among multiple users [4].

The chapter provides a comprehensive description of the implementation environment, controller logic, machine learning integration, QoS mechanisms, handling of encrypted traffic, and system evaluation. The goal is to demonstrate that the system is practical, efficient, and capable of operating in real-world conditions.

4.2 Implementation Environment and Tools

The implementation of the proposed system is carried out using a combination of software tools and networking components that simulate a real smart classroom environment.

The SDN controller is developed using the Ryu framework, which is a Python-based controller that supports OpenFlow protocol for communication with network devices [6]. Ryu is chosen due to its flexibility, modularity, and ease of integration with machine learning models.

Open vSwitch (OVS) is used as the SDN-enabled switch, acting as the data plane of the network. OVS supports OpenFlow and allows the controller to install flow rules dynamically. It is configured to operate in secure mode, ensuring that all unknown packets are forwarded to the controller for processing.

The system is deployed on a Linux-based environment (Ubuntu), where all components including the controller, switch, and machine learning model are executed. Python libraries such as NumPy, Pandas, and Scikit-learn are used for data processing and model inference.

To simulate a classroom environment, multiple client devices (hosts) are connected to the network through a wireless access point. These devices generate different types of traffic, including academic (e-learning platforms, research websites) and non-academic (social media, streaming, gaming).

This setup allows realistic testing and evaluation of the system under controlled conditions.

4.3 SDN Controller Design and Workflow

The SDN controller is the core component of the system, responsible for monitoring network traffic, performing classification, and enforcing QoS policies.

When a packet arrives at the Open vSwitch, the switch checks its flow table for a matching rule. If no rule is found, the packet is encapsulated in a Packet-In message and sent to the controller. This triggers the controller's processing logic.

The controller extracts packet header information, including source and destination IP addresses, port numbers, and protocol type. These attributes are used to identify flows. Each flow is tracked using a unique key, and associated statistics are maintained in a data structure.

The controller continuously updates flow statistics such as packet count, byte count, and timestamps. Once a sufficient number of packets are collected, the controller performs feature extraction and classification.

After classification, the controller installs flow rules in the switch using OpenFlow messages. These rules define how subsequent packets of the flow should be handled, reducing the need for repeated controller involvement.

This reactive-proactive hybrid approach improves efficiency and ensures real-time operation [5].

4.4 Feature Processing and Data Pipeline

The data processing pipeline consists of several stages, including packet capture, flow aggregation, feature extraction, and normalization.

Packets received by the controller are grouped into flows based on a five-tuple (source IP, destination IP, source port, destination port, protocol). For each flow, statistical features are computed dynamically.

Key features include:

- Average packet size
- Flow duration
- Packet inter-arrival time
- Total number of packets
- Total bytes transferred

These features capture the behavioral characteristics of traffic and are effective for distinguishing between application types [3].

To improve classification accuracy, contextual features are also incorporated. DNS responses are analyzed to extract domain names, and TLS Server Name Indication (SNI) is used to identify the target server in encrypted connections.

Before classification, feature values are normalized to ensure consistency and improve model performance. The processed feature vector is then passed to the machine learning model.

4.5 Machine Learning Model Deployment

The machine learning model used in the system is Logistic Regression, which is selected for its efficiency and suitability for real-time applications [7].

The model is trained offline using a labeled dataset of network flows. Each flow is labeled as either academic or non-academic based on its characteristics. The training process involves feature selection, model fitting, and validation.

Once trained, the model is serialized using joblib and loaded into the SDN controller. During runtime, the controller uses the model to classify incoming flows based on extracted features.

The prediction process is computationally lightweight, enabling real-time classification without introducing significant delay. The output of the model is a probability value, which is converted into a binary decision using a threshold.

This integration allows the system to make intelligent decisions dynamically and adapt to changing traffic patterns.

4.6 QoS Policy Implementation

Quality of Service (QoS) policies are implemented to ensure efficient bandwidth allocation and prioritize academic traffic. The system uses two main QoS mechanisms such as DSCP marking and OpenFlow meter tables.

For academic traffic, the controller assigns a high-priority DSCP value in the IP header. This ensures that academic packets are processed preferentially by network devices. For non-academic traffic, rate limiting is applied using OpenFlow meter tables. Each non-academic flow is assigned a bandwidth limit, preventing it from consuming excessive network resources.

The controller dynamically installs QoS rules based on classification results. This enables adaptive bandwidth management and ensures fair resource allocation [8].

4.7 Handling Encrypted and CDN-Based Traffic

Modern network traffic is predominantly encrypted, making payload inspection impossible. The proposed system addresses this challenge by using flow-based and contextual features.

Statistical features provide insights into traffic behavior, while DNS and SNI provide domain-level information. This combination allows the system to classify traffic without accessing payload data.

However, certain limitations exist. Content Delivery Networks (CDNs) cause multiple services to share the same IP address, making classification more complex. DNS-over-HTTPS further reduces visibility into domain information.

Despite these challenges, the system achieves effective classification by combining multiple feature types and machine learning techniques [4].

4.8 Experimental Setup and Test Scenarios

The system is evaluated using a simulated smart classroom environment with multiple connected devices.

Test scenarios include:

- Accessing academic websites (e-learning platforms, research journals)
- Streaming video content
- Browsing social media platforms
- Online gaming and entertainment

Each scenario is designed to test the system's ability to classify traffic and enforce QoS policies.

Performance metrics include:

- Classification accuracy
- Latency
- Bandwidth utilization
- Packet loss

These metrics provide a comprehensive evaluation of system performance.

4.9 Results and Performance Evaluation

The results demonstrate that the system effectively classifies traffic and enforces QoS policies in real time.

Academic traffic is successfully prioritized, resulting in improved performance for educational applications. Non-academic traffic is rate-limited, preventing bandwidth congestion.

The system achieves high classification accuracy due to the use of machine learning and flow-based features. Latency is minimized through efficient controller-switch interaction.

However, certain limitations are observed. Short flows may be misclassified due to insufficient data, and encrypted DNS can reduce classification accuracy.

Overall, the results validate the effectiveness of the proposed system.

4.10 Limitations and Future Improvements

The system has several limitations. The reliance on flow-based features introduces a delay in classification, as sufficient data must be collected before making a decision.

Encrypted DNS and CDN-based traffic reduce visibility and can affect classification accuracy. Additionally, the machine learning model requires periodic retraining to adapt to new traffic patterns.

Future improvements may include the use of deep learning models, integration of additional features, and deployment in larger network environments.

4.11 Chapter Summary

This chapter presented the detailed implementation and evaluation of the proposed system. It demonstrated how SDN and machine learning can be integrated to achieve real-time traffic classification and QoS enforcement.

The results confirm that the system is effective, scalable, and suitable for smart classroom environments.

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